Nahda University in Beni Suef
Faculty of Computer Science
Department of Computer Science

Final Year Student Graduation Project

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Project Concept and Plan

UPPLY

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Project Concept and Plan

Abstract

- ✔ The modern recruitment process is characterized by inefficiencies, primarily stemming from high volumes of applications and subjective initial screening methods. Job seekers often struggle to find relevant openings and experience generic application flows, while recruiters waste significant time sifting through poorly matched candidates. **UPPLY** is proposed as a novel, intelligent job portal that utilizes **Artificial Intelligence (AI)** to streamline these processes and enhance the integrity of the remote screening phase.
- ✔ This project will **Set the Scene** by analyzing the limitations of contemporary Applicant Tracking Systems (ATS), which often rely on simple keyword matching and lack a standardized, objective method for pre-interview evaluation. The **State of the Art** shows various isolated tools for resume analysis and some basic automated testing, but an integrated platform offering comprehensive profile management, intelligent matching, and a fully secured AI-driven assessment remains a research gap.
- ✔ The core **Problem** is two-fold: the low accuracy of automated candidate-to-job matching, and the lack of a scalable, reliable method for testing communication and knowledge while mitigating dishonesty in an online environment. Our **Proposal to Solve the Problem** is the development of **UPLLY**, a robust system featuring two key intelligent components: first, a **profile-based specialization engine** that ensures job seekers are presented with highly relevant job listings tailored to their entered skills and experience; and second, an innovative, integrated **AI-Powered Oral Interview System**. This system will conduct automated, spoken interviews, score linguistic and technical

responses using NLP, and simultaneously utilize **Computer Vision models** for **real-time cheating detection** (proctoring) via the candidate's camera.

✔ The **Aims and Objectives** are to develop a functional platform that significantly reduces the screening time for recruiters (by using an automated sorting model) and provides a secure, objective pre-screening experience. Objectives include the successful development and integration of the profile management, job creation, and application functionalities, and achieving a demonstrable level of accuracy in both the applicant sorting and the cheating detection models. The primary **Contribution** of this thesis will be the design and implementation of the **integrated AI oral assessment and proctoring framework**, offering a new, secure blueprint for scalable remote candidate screening.

Work Breakdown

Phase 1: Research, Definition, and Design (Semester 1)

- **Preliminary Research:** Initial feasibility study and technology assessment.
- **Project Concept (Definition) and Plan:** Finalize project scope, architecture, and schedule.
- **Literature Review:** Detailed research into recruitment systems, NLP, Speech-to-Text models, and Computer Vision for proctoring.
- **Requirements, Analysis, and Design:** Specify all functional and non-functional requirements, database schema, and system architecture.
- **First Semester Thesis Writing:** Draft initial chapters (Introduction, Literature Review, Analysis & Design).
- **Intermediate Progress Report:** Formal submission documenting all Phase 1 work.

Phase 2: Implementation (Semester 1 & 2)

- **Core Platform Implementation:** Develop user registration, profile management, job posting, and application submission modules.
- **AI Matching & Specialization Implementation:** Build the profile-based job display engine and the initial applicant sorting/ranking algorithm.
- **Oral Interview System Implementation:** Develop and integrate the Speech-to-Text, NLP-based scoring, and dialogue management components.
- **Cheating Detection Implementation:** Develop and train the Computer Vision proctoring model and integrate it with the interview module.

Phase 3: Testing, Evaluation, and Documentation (Semester 2)

- **Testing:** Conduct unit, integration, and system testing for all components (platform and AI).
- **Evaluation:** Assess the accuracy and performance of the AI models and the overall system against the thesis aims.
- **Thesis writing:** Completion of remaining chapters, abstract, and final editing.
- **Presentation Preparation:** Create slides and rehearse for the final presentation.

Milestones & Deliverables

Project Concept (Definition) and Plan Deliverable / Milestone
Requirements Analysis Milestone
Software Design Milestone
Intermediate Progress Report Deliverable
Core Platform Functional Milestone
AI Subsystems (Matching, Interview, Proctoring) Milestone
Testing Milestone
Final Thesis Deliverable
Final Presentation Deliverable

The Team: Roles and Responsibilities

State the name of the team members, and who will do what. State the Project Leader / Manager.

Mohamed Hanafy (Project Leader / Backend & Database Developer): Manages project schedule, risks, and communication. Works on **Server-Side Development, API Implementation, and Database Design/Management**. Shares responsibility for the project's data architecture and core backend functionality.

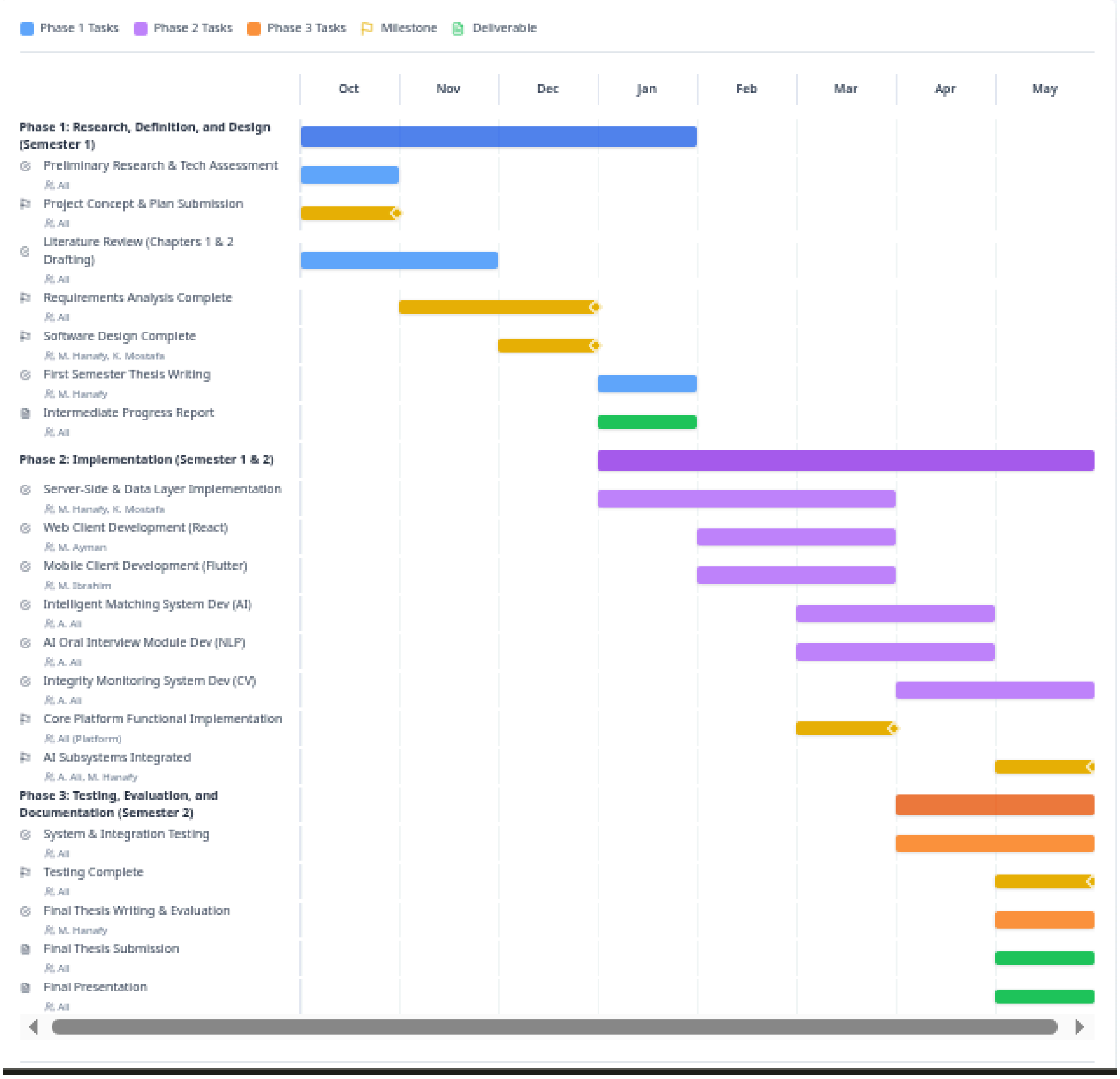
Kareem Mostafa (Backend & Database Developer): Focuses heavily on **Server-Side Development, API Implementation, and Database Design/Management**. Works alongside the Project Lead to build the robust core backend and implement data flow logic.

Mohamed Ibrahim (Mobile Application Lead): Responsible for the entire **Mobile Client Development** and its integration with the core systems, ensuring optimal performance on mobile devices.

Menna Allah Ayman (Web Client Lead): Responsible for all aspects of the **Web Portal Development**, focusing on the user experience (UX) and front-end application.

Aliaa Ali (AI/ML Specialist): Leads the research, development, and deploy of all **Artificial Intelligence** modules, including predictive models and real-time computer vision systems.

Gantt chart


Phase 1 Tasks
Phase 2 Tasks
Phase 3 Tasks
Milestone
Deliverable
Oct
Nov
Dec
Jan
Feb
Mar
Apr
May
Phase 1: Research, Definition, and Design (Semester 1)
Preliminary Research & Tech Assessment
All
Project Concept & Plan Submission
All
Literature Review (Chapters 1 & 2 Drafting)
All
Requirements Analysis Complete
All
Software Design Complete
M. Hanafy, K. Mostafa
First Semester Thesis Writing
M. Hanafy
Intermediate Progress Report
All
Phase 2: Implementation (Semester 1 & 2)
Server-Side & Data Layer Implementation
M. Hanafy, K. Mostafa
Web Client Development (React)
M. Ayman
Mobile Client Development (Flutter)
M. Ibrahim
Intelligent Matching System Dev (AI)
A. Ali
AI Oral Interview Module Dev (NLP)
A. Ali
Integrity Monitoring System Dev (CV)
A. Ali
Core Platform Functional Implementation
All (Platform)
AI Subsystems Integrated
A. Ali, M. Hanafy
Phase 3: Testing, Evaluation, and Documentation (Semester 2)
System & Integration Testing
All
Testing Complete
All
Final Thesis Writing & Evaluation
M. Hanafy
Final Thesis Submission
All
Final Presentation
All